

Curriculum Vitae

Harry Stoltz

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Education

University of California, Riverside (Cum Laude) <i>Bachelor of Science in Pure Mathematics; GPA: 3.87</i>	Riverside, CA Sep. 2021 – Jun. 2025
New York University, Courant Institute <i>Master's Student, Mathematics</i> Reading course with Professor Hisham Sati on categorical quantum mechanics and higher mathematical structures.	New York City, NY Sep. 2025 – Present

Research Interests

Categorical quantum mechanics, ZX-calculus, stabilizer quantum theory, diagrammatic reasoning, compositional quantum software, and applied category theory.

Research Experience

Graduate Researcher – ZX-Calculus and CQM <i>New York University, Courant Institute</i> - Resolved the two remaining rule-independence cases in the Backens–Perdrix–Wang minimality program for the stabilizer ZX-calculus, proving that the simplified rule set has no redundant rewrite rule. - Authored <i>The Simplified Stabilizer ZX-Calculus is Minimal</i>, arXiv:2606.12383 [quant-ph]. - Presented the stabilizer minimality work at the 2026 Mathematical Pathways to an Excellent Future Conference. - Accepted for a poster presentation on the stabilizer minimality work at QPL 2026 (Quantum Physics and Logic). - Developed ZX Sketch (zxsketch.com), an online ZX diagram editor with a complete rule set for supported workflows and PyZX as the underlying engine.	Sep. 2025 – Present New York City, NY
SURF Student <i>California Institute of Technology</i> - Researched 2-D quantum tori and Hilbert's 12th problem with Professor Matilde Marcolli. - Presented research in October 2024 at SURF conference.	Jun. 2024 – Sep. 2024 Pasadena, CA
Undergraduate Student Researcher <i>University of California, Riverside</i> - Researched genetic mosaicism using novel stem cell technology with Professor Thomas E. Kuhlman. - Continued research as part of UCR's prestigious summer RISE program during Summer 2023. - Presented research at the RISE symposium to 200+ researchers. - Co-authored MosaicLev paper on mobile element-aware genome comparison (bioRxiv:2026.02.03.703520).	Apr. 2023 – Jun. 2025 Riverside, CA
Undergraduate Student Researcher <i>University of California, Riverside</i> - Researched public health interventions in various African countries with Professor Kim Yi Dionne. - Published a paper on the efficacy of undergraduate student research in humanities.	Dec. 2021 – Jun. 2025 Riverside, CA

Software Projects

- ZX Sketch** | *TypeScript, Pyodide/WebAssembly* Oct. 2025 – Present
- Online ZX diagram editor for constructing, rewriting, and simplifying ZX-calculus diagrams in the browser.
 - Supports a complete rule set for supported workflows, proof mode, branching, and goal verification.
 - Uses PyZX as the underlying simplification engine via Pyodide/WebAssembly.
 - Exports TikZ, SVG, and PNG; imports PyZX and ZXLive-style formats; deployed at zxsketch.com.
 - Public source code: github.com/hstoltz/ZX-sketch.

Teaching Experience

- Teaching Assistant: New York University** Sep. 2025 – Present
- TA for Calculus II with Professor Matthew Leingang (Fall 2025, Spring 2026).
- Student Instructor: University of California, Riverside** Jan. 2023 – Jun. 2025
- Designed, created, and implemented a new course at UCR: “International Relations as a Function of Spaceflight.”
 - Worked with UCR Faculty (Professor Kim Dionne) to set up the class, and took EDUC 102 “Democratic Pedagogy: Developing R’Courses” in preparation.
 - Taught 8 quarters: Winter 2023 through Spring 2025.

Science Communication & Writing Experience

- Science Communications Intern: California Institute of Technology** Jun. 2022 – Sep. 2022
- Wrote articles on novel research at Caltech.
 - Interviewed Caltech staff about their research.
- Freelance Writer: Inverse.com; Bustle Media Group** Mar. 2022 – Present
- Occasionally commissioned to write articles on current news in spaceflight.
- Staff Writer: The Highlander** Jan. 2022 – Sep. 2023
- Published articles weekly in the student newspaper as a paid writer.
- Journalist: Space Explored** Aug. 2021 – Present
- Published articles as a paid journalist.
- Lead Space Correspondent: CleanTechnica** Sep. 2019 – Present
- Published articles as a paid journalist, attended NASA press events, and traveled to SpaceX press event.

Leadership & Service

- R’Course Review Board: University of California, Riverside** Nov. 2024 – Jun. 2025
- Read, review, and revise R’course course proposals.
- Pacific Math Alliance VP: University of California, Riverside** Sep. 2024 – Jun. 2025
- Club planning, outreach, tabling, and lead seminars on various math topics, including my popular “Desmos workshop,” offered quarterly.
- Math Tutoring** 2022 – Present
- Tutoring students – primarily in Calculus, Linear Algebra, and Differential Equations.
- Community Outreach via PMA** 2025
- Initiated community outreach program in the Riverside area.
 - Used position as Pacific Math Alliance VP to help underserved high school students get excited about mathematics.

Publications

- **The Simplified Stabilizer ZX-Calculus is Minimal.** Stoltz, H. K. [arXiv:2606.12383](https://arxiv.org/abs/2606.12383) [quant-ph], 2026.
- **ZX Sketch: An Online Graphical Editor for the ZX Calculus.** In preparation.
- **MosaicLev: Modified Levenshtein distance for mobile element-aware genome comparison.** Stoltz, H. K. & Kuhlman, T. E. [bioRxiv:2026.02.03.703520](https://doi.org/10.1101/2026.02.03.703520), 2026.
- **A Mathematical Model for Genetic Mosaicism.** UCR Honors Capstone, 2024–2025.
- **Undergraduates and Political Science Research: Insights from Research Assistants in a Minority-Serving Institution Lab:** 2023
Velásquez, Zabdi R., Jasmine Esmail, Harry Stoltz, Fulya Felicity Turkmen, Owura Kuffuor, John Burnett, Naia Pizarro et al. "Undergraduates and Political Science Research: Insights from Research Assistants in a Minority-Serving Institution Lab." PS: Political Science & Politics 56, no. 4 (2023): 493-498.

Conference Presentations & Participation

- **The Simplified Stabilizer ZX-Calculus is Minimal – QPL 2026 Poster Presentation:** Aug. 2026 (accepted)
- **The Simplified Stabilizer ZX-Calculus is Minimal – 2026 Mathematical Pathways to an Excellent Future Conference:** May 2026
- **Science x AI Summit 2026 – SAIR Foundation, UC Riverside:** Jun. 29–30, 2026 (upcoming)
- **A Mathematical Model for Genetic Mosaicism – 2025 UCR Undergraduate Research & Creative Activities Symposium:** Apr. 2025
- **A Mathematical Model for Genetic Mosaicism – Mathematical Physics Seminar:** Apr. 2025
- **Geodesics Between Thetas – 2025 Computational Interactions between Algebra, Combinatorics, and Discrete Geometry:** Feb. 2025
- **Geodesics Between Thetas – 2024 SURF Fall Seminar Day:** Sep. 2024
- **Formalization of Euclidean Geometry via Hilbert's Axioms – Mathematical Pathways to an Excellent Future Conference:** May 2024
- **Formalization of Euclidean Geometry via Hilbert's Axioms – 2024 UCR Undergraduate Research & Creative Activities Symposium:** Apr. 2024
- **Quantitatively Measuring the Effects of Genetic Mosaicism – 2023 RISE/MARC Symposium:** Sep. 2023

Honors and Awards

UCR Honors Student (Sep. 2021 – Jun. 2025)

- Took Honors-exclusive courses at UCR.
- Wrote and published thesis: "A Mathematical Model for Genetic Mosaicism."

Mathematics Honors Distinction (2021–2025)

Student on the Dean's Honors List (2021–2025, quarterly)

- Maintained a minimum 3.5 GPA per quarter.

Student on the Chancellor's Honors List (2021–2025, annually)

- Qualified by being on the Dean's List every quarter of the academic year.

Cum Laude (2025)